

SAINT-MALO-DINARD AP, France

WMO: 071250

Lat: 48.5889N

Lon: 2.0748W

Elev: 67

StdP: 100.53

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) -0.8	(d) -6.0	(e) 2.3	(f) -0.5	(g) -4.2	(h) 2.7	(i) 0.7	(j) 13.6	(k) 10.2	(l) 12.4	(m) 10.3	(n) 2.8	(o) 90	(p) 0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 26.9	(d) 19.6	(e) 24.6	(f) 18.6	(g) 22.9	(h) 17.7	(i) 20.3	(j) 25.3	(k) 19.3	(l) 23.3	(m) 18.4	(n) 21.9	(o) 4.0	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	22.6	17.7	12.8	21.2	16.9	12.2	20.3	58.5	25.4	55.1	23.6	52.2	21.9	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.7	8.7	-5.3	31.9	1.7	2.9	-6.5	34.0	-7.5	35.7	-8.4	37.3	-9.7	39.4
			-5.7	22.5	1.7	1.5	-7.0	23.5	-8.0	24.4	-9.0	25.2	-10.3	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	6.4	6.6	8.3	10.2	13.0	15.9	17.6	17.4	16.0	13.1	9.4	7.0
	DBStd	4.96	3.42	3.14	2.91	2.70	2.50	2.63	2.21	2.20	2.51	2.81	2.74	3.43
	HDD10.0	469	117	101	69	28	3	0	0	0	0	6	42	103
	HDD18.3	2448	370	329	311	243	167	83	41	43	78	164	267	352
	CDD10.0	1113	5	5	16	35	96	176	234	230	181	102	25	9
	CDD18.3	51	0	0	0	0	1	9	17	14	8	1	0	0
	CDH23.3	341	0	0	0	2	14	79	108	86	48	4	0	0
	CDH26.7	78	0	0	0	0	2	21	26	21	8	1	0	0
(14) Wind	WSAvg	4.4	5.0	4.9	4.7	4.4	4.2	3.9	4.0	3.9	3.9	4.2	4.7	4.9
(15) Precipitation	PrecAvg	723	65	58	46	53	53	48	47	53	55	77	89	78
	PrecMax	877	132	129	143	126	120	102	98	130	147	152	167	181
	PrecMin	552	8	11	11	7	13	12	18	10	7	16	24	15
	PrecStd	94	37	30	30	28	26	26	23	32	30	32	38	37
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.2	19.1	22.9	25.2	29.7	29.8	29.9	27.8	22.5	17.5	14.7
		MCWB	12.4	11.5	12.8	14.4	18.4	21.4	20.7	21.3	19.3	17.6	15.3	13.0
	2%	DB	13.3	13.3	16.4	19.5	21.7	25.3	26.0	25.6	24.4	20.0	15.8	13.8
		MCWB	11.6	10.8	11.7	13.9	16.1	19.1	19.1	19.2	18.4	16.4	13.9	12.3
	5%	DB	12.2	12.2	14.4	16.9	19.6	22.8	23.9	23.3	22.2	18.4	14.6	12.9
		MCWB	10.7	10.2	11.2	12.6	15.0	17.7	18.2	18.3	17.4	15.6	12.7	11.5
	10%	DB	11.2	11.2	13.0	14.9	17.8	20.9	22.2	21.8	20.6	17.2	13.5	11.9
		MCWB	9.8	9.5	10.5	11.2	14.1	16.5	17.5	17.3	16.7	15.0	11.9	10.5
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	12.4	13.7	16.2	18.9	22.0	21.7	21.5	20.8	18.4	15.9	13.4
		MCDB	14.1	14.4	17.4	20.8	24.4	27.0	28.5	28.5	25.5	21.2	17.0	14.4
	2%	WB	11.9	11.4	12.5	14.1	16.7	19.7	20.0	19.9	19.2	17.0	14.1	12.5
		MCDB	13.2	12.9	15.1	18.7	20.6	24.7	24.5	24.1	22.9	19.1	15.5	13.6
	5%	WB	10.9	10.5	11.6	12.9	15.5	18.1	18.9	18.8	18.1	16.2	13.1	11.7
		MCDB	12.0	12.0	14.0	16.3	18.8	21.9	22.6	22.3	21.2	17.9	14.3	12.8
	10%	WB	10.0	9.6	10.7	11.7	14.4	16.8	18.0	18.0	17.2	15.3	12.2	10.7
		MCDB	11.1	11.0	12.8	14.3	17.2	20.2	21.4	20.9	19.8	16.9	13.3	11.8
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	7.2	8.2	8.1	8.5	8.6	8.5	8.6	6.9	5.8	5.3
		MCDBR	5.4	7.0	9.5	11.8	11.4	12.1	11.9	11.3	10.9	7.7	6.4	5.0
	5% WB	MCWBR	4.9	5.2	6.0	6.8	6.6	6.7	6.2	5.7	5.8	4.8	4.9	4.5
		MCDWR	5.2	6.4	8.4	10.5	10.3	11.1	10.4	9.8	9.5	7.0	6.0	5.1
(42) Clear-Sky Solar Irradiance	taub		0.329	0.345	0.374	0.389	0.395	0.392	0.389	0.380	0.365	0.358	0.334	0.322
		taud	2.424	2.394	2.314	2.288	2.315	2.356	2.365	2.413	2.434	2.449	2.442	2.396
	Ebn at Noon		731	804	834	859	866	870	867	860	837	779	717	682
		Edh at Noon	67	85	107	121	123	119	116	106	94	80	65	60
(44) All-Sky Solar Radiation	RadAvg		0.97	1.75	2.83	4.34	5.14	5.67	5.42	4.62	3.65	2.10	1.18	0.81
	RadStd		0.09	0.20	0.31	0.45	0.43	0.62	0.35	0.38	0.28	0.17	0.12	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page